

Email Phishing Detection Using Machine Learning

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Course:BSc IT Academic Year:2025-26

❖ INTRODUCTION

- Email phishing is a major cybersecurity threat.
- Attackers trick users into revealing sensitive information.
- Manual detection is difficult and time-consuming.
- Machine Learning helps automate phishing detection.

❖ PROBLEM STATEMENT

- Increase in phishing attacks via emails.
- Users find it hard to identify fake emails.
- Traditional security methods are not always effective.
- Need for an automated phishing detection system.

❖ OBJECTIVES

- To detect phishing emails automatically.
- To classify emails as Phishing or Legitimate.
- To use machine learning for accurate prediction.
- To provide a simple command-line interface

❖ DATASET DESCRIPTION

- Dataset name: emails.csv
- Contains phishing and legitimate emails.
- Stored in the dataset folder.
- Used for training and testing the model.

❖ TECHNOLOGIES USED

Programming Language: Python

- Libraries:
- Pandas
- NumPy
- Scikit-learn
- Concepts:
- Machine Learning
- Natural Language Processing (NLP)
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❖ SYSTEM ARCHITECTURE

- Input Email Text
- Text Preprocessing
- Feature Extraction
- Machine Learning Model
- Prediction Output

❖ WORKING OF THE SYSTEM

Dataset is loaded and cleaned.

- Text is converted into numerical features.
- Machine learning model is trained.
- User enters an email.
- System predicts phishing or safe email.
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❖ MODULES USED

- preprocess.py – cleans email text
- model.py – handles model training
- phishing_detection.py – main execution file

❖ OUTPUT SCREENSHOTS

- Model trained successfully
- User enters email
- Prediction result shown

```
(venv) C:\Users\janga\OneDrive\Desktop\Email_  
Model Accuracy: 0.0  
[E] This email is Safe!
```

❖ ADVANTAGES

- Fast and automated detection
- Reduces risk of cyber fraud
- Easy to use
- Improves email security

❖ LIMITATIONS

- Accuracy depends on dataset quality
- Cannot detect completely new phishing patterns
- Command-line based interface

❖ FUTURE SCOPE

- Web-based application Real-time
- email scanning Deep learning
- integration Browser or email
- client extension

❖ CONCLUSION

- Email phishing is a serious security threat.
- Machine learning helps detect phishing effectively.
- The system successfully classified emails.
- Project achieves its objectives.